



RT58x Bootloader Guide

V1.0

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1. Introduction

This document describes RT58x bootloader flow.

2. Bootloader

此章節主要在說明 bootloader 的功能與流程。

2.1 Feature

- Support upgrade firmware
- Support decompression (LZMA with a 64KB (2^{16} bytes) dictionary size)
- Support decryption

2.2 Flow

Bootloader 可以更新 firmware，更新檔案支援壓縮與 Security。在進行更新前會去位置 0x7000(Upgrade Information Address)取得相關資訊，才能夠進行更新。最後跳到 0x8000(Firmware Address)的位置執行 firmware。

流程如圖 1 所示：

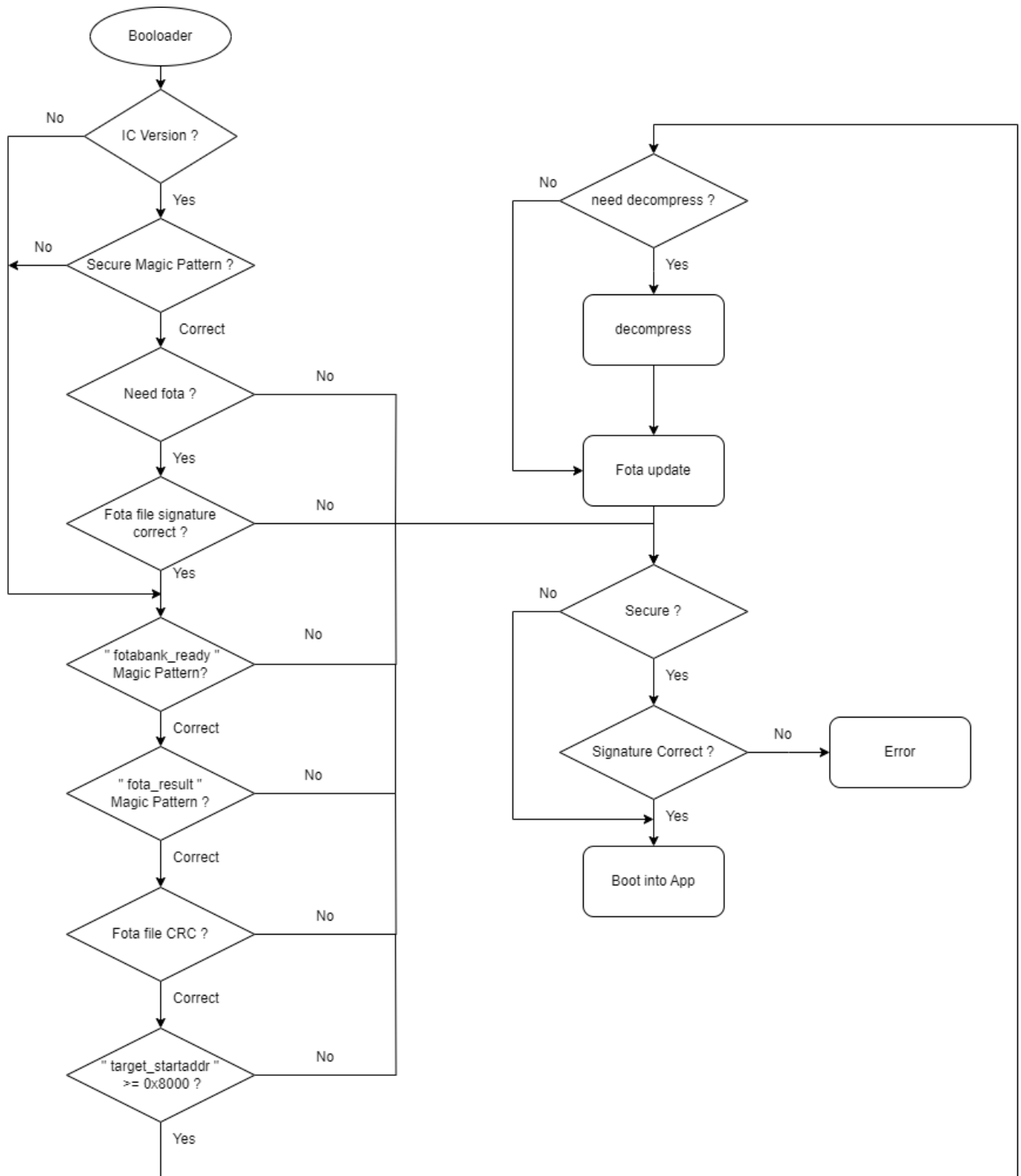


Figure1. Bootloader flow chat

首先檢查 IC 的板是否支援 security，若有支援則檢查 security 的 Magic Pattern(0x72A34B50)，再來檢查是否有要更新。若沒有要更新，則去進行簽章確認。如果有要更新就會去 Upgrade Information Address 取得資訊。

再來檢驗 fota file 的簽章是否正確，檢查 fotabank_ready 是否為 Magic Pattern (0xA55A6543)、fota_result 是否有被寫過，最後檢查 CRC 和 target_startaddr 位置。若檔案是壓縮過的，則進行解壓，之後進行更新，更新的結果最終會寫在 0x7018，0x00 代表成功，非 0x00 和非 0xFF 的表示失敗。

更新完之後會檢查更新後的簽章是否正確，若正確則開始跑 firmware。

若在更新前有發生問題則會跑舊有的程式，仍可以正常開機運作，重新做 fota。

3. Upgrade Information

此章節描述 Upgrade Information Address 的資料。

為了要進行 upgrade firmware，需要在 Upgrade Information Address 獲得的資訊如下

Address	Octet	Name	Field Description	Value
0x7004	4	fotabank_ready	Binary image ready magic number	0xA55A6543
0x7008	4	fotabank_crc	Binary Image file CRC value	CRC value
0x700C	4	fotabank_data_len	Binary image file length	Length value
0x7010	4	fotabank_startaddr	Binary image file start address	Start address
0x7014	4	target_startaddr	Application start address	0x8000
0x7019	1	fota_image_info	Image information	Information
0x701A	4	compressed_len	Compressed length ^{※1}	Compressed length

※1 This parameter is only used in “Compression & Security” Mode. This parameter indicates the length after signature and compression, which is equivalent to the “Compressed file length” in RT58x Image Upgrade flow & FOTA Image Structure Guide Chapter 8.1.1 (please force the conversion of 3 bytes to 4 bytes).

各欄位說明如下

fotabank_ready :

當值為 0xA55A6543，表示可以更新

fotabank_crc :

檔案的 CRC，確保檔案在傳輸過程中沒有出錯

fotabank_data_len :

檔案長度，單位(Byte)

fotabank_startaddr :

更新檔案存放的地址，若有壓縮只能寫在下列特定位置

RT582 : 0x99000

RT583 : 0x103000

target_startaddr :

要複寫的位置，主程式會在0x8000的位置

fota_image_info :

bit field來攜帶資訊，0x01是壓縮過，0x02是有簽章

compressed_len :

此用在檔案有壓縮加上簽章時要填入，為壓縮過後的檔案長度。

4. Upgrade Information Example

此章節描述bootloader要進行更新firmware時，預期會獲得upgrade information的數據。

4.1 Normal

fotabank_ready : 0xA55A6543

fotabank_crc : CRC

fotabank_datalen : fota file length

fotabank_startaddr : fota file storage address

target_startaddr : 0x8000

fota_image_info : 0x00

compressed_len : N/A

4.2 Compress

fotabank_ready : 0xA55A6543

fotabank_crc : CRC

fotabank_datalen : length

fotabank_startaddr : address

target_startaddr : 0x8000

fota_image_info : 0x01

compressed_len : N/A

4.3 Security

fotabank_ready : 0xA55A6543

fotabank_crc : CRC

fotabank_data_len : length

fotabank_start_addr : address

target_start_addr : 0x8000

fota_image_info : 0x02

signature_len : N/A

4.4 Compress and Security

fotabank_ready : 0xA55A6543

fotabank_crc : CRC

fotabank_data_len : length

fotabank_start_addr : address

target_start_addr : 0x8000

fota_image_info : 0x03

compressed_len : compress length

Revision History

Revision	Description	Owner	Date
1.0	Initial version	KC Tseng	2023/10/19

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